

Task Description:

Create a custom docker image for nginx and deploy it using docker compose, where the volume bind mount should be at /var/opt/nginx location. Push the created custom docker image to your docker-hub.

Output:

Instance summary for i-Ofa59a354341849af (task3) Info

Updated less than a minute ago

Instance ID i-Ofa59a354341849af	Public IPv4 address 13.233.114.77 open address	Private IPv4 addresses 172.31.37.51
IPv6 address -	Instance state Running	Public DNS ec2-13-233-114-77.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-37-51.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-37-51.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPV4 (A)	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation s . Learn more
Auto-assigned IP address 13.233.114.77 [Public IP]	VPC ID vpc-0421d1e9b2d53db74	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0325f240738e9ee70	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:ap-south-1:177593955230:instance/i-Ofa59a354341849af	

```
Setting up runc (1.3.3-0ubuntu1-24.04.3) ...
Setting up dns-root-data (2024071801~ubuntu0.24.04.1) ...
Setting up bridge-utils (1.7.1-1ubuntu2) ...
Setting up pigz (2.8-1) ...
Setting up containerd (1.7.28-0ubuntu1-24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.service.
Setting up ubuntu-fan (0.12.16+24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /usr/lib/systemd/system/ubuntu-fan.service.
Setting up docker.io (28.2.2-0ubuntu1-24.04.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group 'docker' (GID 113) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-37-51:~$
```

i-Ofa59a354341849af (task3)

PublicIPs: 13.233.114.77 PrivateIPs: 172.31.37.51

```
aws | Search [Alt+S] | Lakshmi Srinath (1775-9395-5230) | Lakshmi Srinath

Setting up containerd (1.7.28-0ubuntu1~24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.service.
Setting up ubuntu-fan (0.12.16+24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /usr/lib/systemd/system/ubuntu-fan.service.
Setting up docker.io (28.2.2-0ubuntu1~24.04.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group 'docker' (GID 113) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-37-51:~$ sudo systemctl start docker
ubuntu@ip-172-31-37-51:~$ sudo systemctl enable docker
ubuntu@ip-172-31-37-51:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
ubuntu@ip-172-31-37-51:~$
```

```
aws | Search [Alt+S] | Lakshmi Srinath (1775-9395-5230) | Lakshmi Srinath

Setting up docker.io (28.2.2-0ubuntu1~24.04.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group 'docker' (GID 113) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

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No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-37-51:~$ sudo systemctl start docker
ubuntu@ip-172-31-37-51:~$ sudo systemctl enable docker
ubuntu@ip-172-31-37-51:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
ubuntu@ip-172-31-37-51:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-37-51:~$ newgrp docker
ubuntu@ip-172-31-37-51:~$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
ubuntu@ip-172-31-37-51:~$
```

```
aws | Search [Alt+S] | Lakshmi Srinath (1775-9395-5230) | Lakshmi Srinath

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-37-51:~$ sudo systemctl start docker
ubuntu@ip-172-31-37-51:~$ sudo systemctl enable docker
ubuntu@ip-172-31-37-51:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
ubuntu@ip-172-31-37-51:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-37-51:~$ newgrp docker
ubuntu@ip-172-31-37-51:~$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
ubuntu@ip-172-31-37-51:~$ sudo curl -L "https://github.com/docker/compose/releases/download/v2.24.6/docker-compose-$(uname -s)-$(uname -m)" \
-o /usr/local/bin/docker-compose
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0    0    0    0     0      0  0 --:--:-- --:--:-- --:--:--    0
100 58.5M  100 58.5M    0     0 24.2M      0  0:00:02  0:00:02 --:--:-- 29.4M
ubuntu@ip-172-31-37-51:~$ sudo chmod +x /usr/local/bin/docker-compose
ubuntu@ip-172-31-37-51:~$ docker-compose --version
Docker Compose version v2.24.6
ubuntu@ip-172-31-37-51:~$
```

```
aws  [+] Search [Alt+S] Asia Pacific (Mumbai) Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-37-51:~$ sudo systemctl start docker
ubuntu@ip-172-31-37-51:~$ sudo systemctl enable docker
ubuntu@ip-172-31-37-51:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
ubuntu@ip-172-31-37-51:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-37-51:~$ newgrp docker
ubuntu@ip-172-31-37-51:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
ubuntu@ip-172-31-37-51:~$ sudo curl -L "https://github.com/docker/compose/releases/download/v2.24.6/docker-compose-$(uname -s)-$(uname -m)" \
-o /usr/local/bin/docker-compose
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0    0    0    0    0    0     0     0  --:--:-- --:--:-- --:--:--    0
100 58.5M 100 58.5M    0    0  24.2M    0  0:00:02  0:00:02 --:--:-- 29.4M
ubuntu@ip-172-31-37-51:~$ sudo chmod +x /usr/local/bin/docker-compose
ubuntu@ip-172-31-37-51:~$ docker-compose --version
Docker Compose version v2.24.6
ubuntu@ip-172-31-37-51:~$ mkdir nginx-docker
ubuntu@ip-172-31-37-51:~$ cd nginx-docker
ubuntu@ip-172-31-37-51:~/nginx-docker$ mkdir html nginx
ubuntu@ip-172-31-37-51:~/nginx-docker$ nano html/index.html
ubuntu@ip-172-31-37-51:~/nginx-docker$ nano nginx/default.conf
ubuntu@ip-172-31-37-51:~/nginx-docker$ nano Dockerfile
ubuntu@ip-172-31-37-51:~/nginx-docker$ nano docker-compose.yml
ubuntu@ip-172-31-37-51:~/nginx-docker$ docker-compose up -d --build
[+] Building 7.8s (8/8) FINISHED
docker:default
```

A screenshot of a web browser showing the Docker Device Activation page. The browser's address bar at the top displays the URL 'login.docker.com/activate'. The page itself has a light gray background. In the center, there is a white rectangular box containing the Docker logo (a blue ship icon) and the word 'docker' in blue. Below the logo, the text 'Device Activation' is displayed in a bold, black font. Underneath that, a smaller line of text reads 'Enter the code displayed on your device'. There is a white input field with a thin gray border containing the placeholder text 'Enter your one-time code *'. At the bottom of the white box is a solid blue button with the word 'Continue' written in white text.

```
aws [Search] [Alt+S] Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath

USING WEB-BASED LOGIN
Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: WQHR-WFBK
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
^Clogin canceled
ubuntu@ip-172-31-37-51:~/nginx-docker$ docker login

USING WEB-BASED LOGIN
Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: ZP2D-NCFB
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-172-31-37-51:~/nginx-docker$
```

```
aws [Search] [Alt+S] Ask Amazon Q Lakshmi Srinath (1775-9395-5230) Lakshmi Srinath

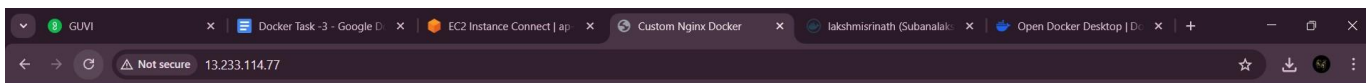
=> => exporting layers 0.0s
=> => writing image sha256:3a087558c110e1e2d2acbl6147ec148f0ba6d0f4647cb23769ccbafda99d2473 0.0s
=> => naming to docker.io/lakshmisrinath/custom-nginx:1.0 0.0s
[+] Running 1/1 0.4s
✔ Container custom-nginx Started

ubuntu@ip-172-31-37-51:~/nginx-docker$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
f6475d682c41  lakshmisrinath/custom-nginx:1.0    "/docker-entrypoint..." 29 seconds ago Up 29 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp custom-nginx

ubuntu@ip-172-31-37-51:~/nginx-docker$ docker login
Authenticating with existing credentials... [Username: lakshmisrinath]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
ubuntu@ip-172-31-37-51:~/nginx-docker$ docker push lakshmisrinath/custom-nginx:1.0
The push refers to repository [docker.io/lakshmisrinath/custom-nginx]
380b77d9b17b: Pushed
7cd9e32fcae2: Pushed
d9d3f8c27ad7: Pushed
4b53e01dba29: Pushed
3b4fce0e490d: Pushed
4c34f6878173: Pushed
547c913b4108: Pushed
e84c0e25063e: Pushed
e50a58335e13: Pushed
1.0: digest: sha256:116f5804b21999566003df01369a88d23130a12155ecb1eba9654a1555ebf4da size: 2192
ubuntu@ip-172-31-37-51:~/nginx-docker$
```



Nginx running via Docker Compose

Docker task 3 done